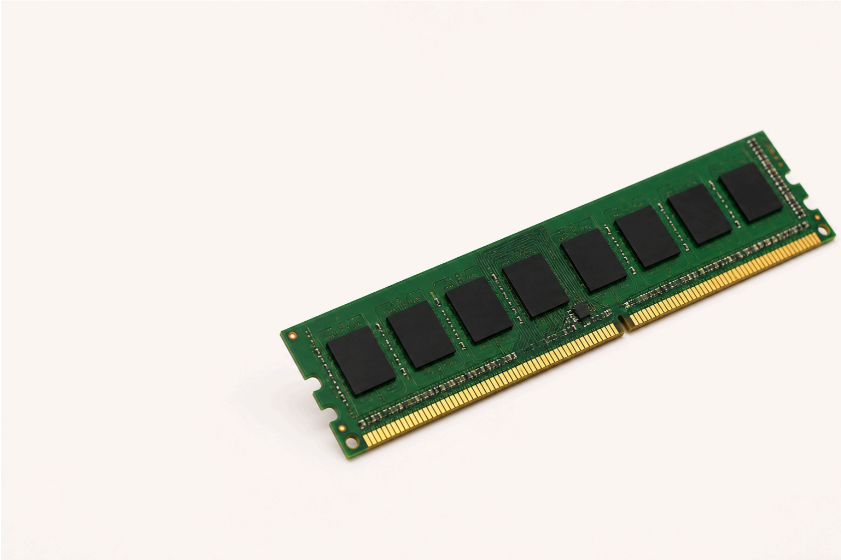
Lecture 2: Data Representation I – Bits, Bytes & Integers

Session 2 · Mon Aug 31, 2026

From Physical Memory to One Byte

That is a **RAM module** — physical hardware, many chips, not one byte.

Memory is **byte-addressable**: every address names one **8-bit byte**, and a byte is the smallest unit a C program can address on its own.



One byte = 8 bits. Today is about what fits in one, and how to work on it.

One Byte, Two Readings

Start with one byte: eight switches that are each either 0 or 1.

One byte: the bit pattern for 65

0	1	0	0	0	0	0	1
128	64	32	16	8	4	2	1

64 + 1 = 65

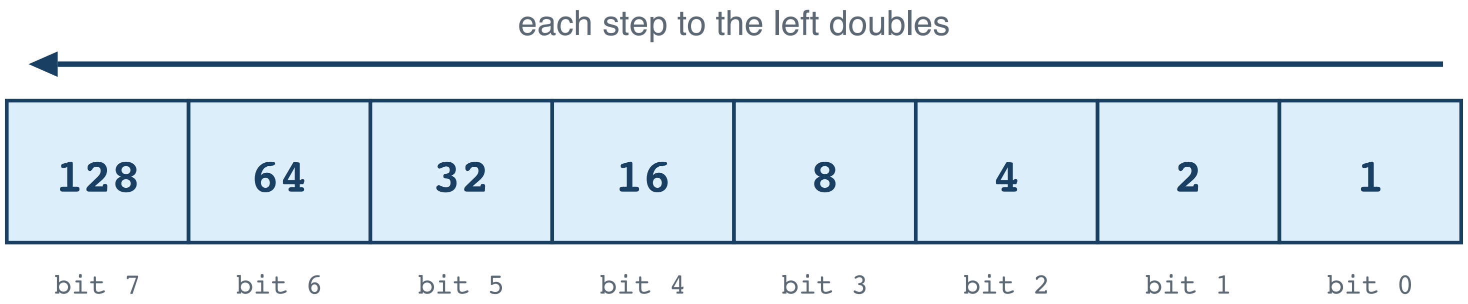
as text (ASCII): A

The computer stores only the bits; the program decides how to read them.

Counting in Binary

Why does `01000001` mean 65? Because each place is worth twice the one to its right — the same idea as decimal's ones, tens and hundreds, with 2 instead of 10.

Each place is worth twice the place to its right



Turn a bit on and you add its place value.

128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255

Counting Up

Counting works exactly the way it does in decimal — when a column runs out of digits, it rolls over and carries into the next one:

```
0 = 0000    3 = 0011    6 = 0110    9  = 1001
1 = 0001    4 = 0100    7 = 0111    10 = 1010
2 = 0010    5 = 0101    8 = 1000    11 = 1011
```

Each extra bit doubles the patterns available: 8 bits give **2^8 = 256**. So an unsigned byte runs 0–255, and an N-bit unsigned value runs 0 to `2^N - 1` — its **UMax**, modulo `2^N`.

Writing the Same Number Three Ways

A number does not change when you write it differently. C lets you say which notation you are using with a **prefix**:

Written	Prefix means	Value
`42`	no prefix — plain decimal	forty-two
`0b101010`	`0b` — read the digits as **binary**	forty-two
`0x2A`	`0x` — read the digits as **hexadecimal**	forty-two

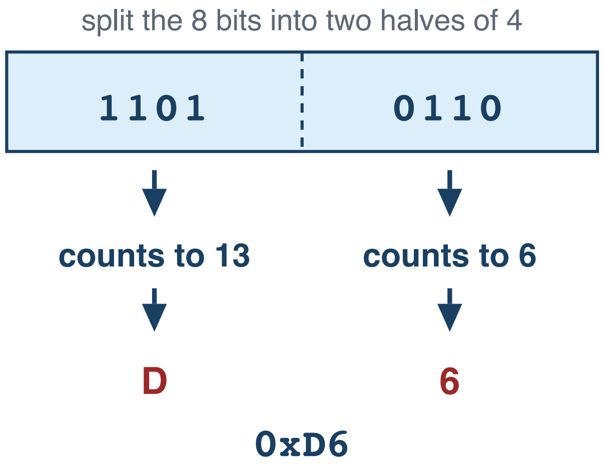
All three are the same value to the compiler. You pick whichever makes the thing you are doing readable: decimal for counting, binary for seeing bits, hex for writing bytes compactly.

> **Note**
>
> `0b` is not standard until C23. GCC and Clang accept it, and these slides use it to show bit patterns, but in ISO C17 write `0x2A` or `42`.

Why Hex Fits Bytes So Well

Eight ones and zeros is easy to miscount. Four bits count 0–15 — exactly one hex digit — so one byte is always two digits, and you can still read the bits straight off the page.

One byte = two hex digits



4 bits count 0 to 15,
so we need 6 extra digits:

```
10 = A
11 = B
12 = C
13 = D
14 = E
15 = F
```

0 through 9 stay as they are.

Characters Are Just Numbers

The rule that turns 65 into `A` is **ASCII** — the American Standard Code for Information Interchange — a table agreed in the 1960s that gives every character a number from 0 to 127.

Characters	Codes	Worth knowing
`'0'`–`'9'`	48–57 (`0x30`–`0x39`)	consecutive, so `c - '0'` gives the digit
`'A'`–`'Z'`	65–90 (`0x41`–`0x5A`)	consecutive
`'a'`–`'z'`	97–122 (`0x61`–`0x7A`)	exactly 32 more than the capitals

```
char c = '7';
int  d = c - '0';                          // 7
int  is_digit = (c >= '0' && c <= '9');   // 1
```

In C the literal `'A'` **is** the number 65. Nothing clever is happening.

The Whole Table

ASCII in full — every code from 0 to 127

Read down a column. Grey are control codes with no glyph; the tinted blocks are the digits, the capitals and the lowercase letters.

0	00	NUL	16	10	DLE	32	20	SP	48	30	0	64	40	@	80	50	P	96	60	`	112	70	p
1	01	SOH	17	11	DC1	33	21	!	49	31	1	65	41	A	81	51	Q	97	61	a	113	71	q
2	02	STX	18	12	DC2	34	22	"	50	32	2	66	42	B	82	52	R	98	62	b	114	72	r
3	03	ETX	19	13	DC3	35	23	#	51	33	3	67	43	C	83	53	S	99	63	c	115	73	s
4	04	EOT	20	14	DC4	36	24	$	52	34	4	68	44	D	84	54	T	100	64	d	116	74	t
5	05	ENQ	21	15	NAK	37	25	%	53	35	5	69	45	E	85	55	U	101	65	e	117	75	u
6	06	ACK	22	16	SYN	38	26	&	54	36	6	70	46	F	86	56	V	102	66	f	118	76	v
7	07	BEL	23	17	ETB	39	27	'	55	37	7	71	47	G	87	57	W	103	67	g	119	77	w
8	08	BS	24	18	CAN	40	28	(	56	38	8	72	48	H	88	58	X	104	68	h	120	78	x
9	09	HT	25	19	EM	41	29	)	57	39	9	73	49	I	89	59	Y	105	69	i	121	79	y
10	0A	LF	26	1A	SUB	42	2A	*	58	3A	:	74	4A	J	90	5A	Z	106	6A	j	122	7A	z
11	0B	VT	27	1B	ESC	43	2B	+	59	3B	;	75	4B	K	91	5B	[	107	6B	k	123	7B	{
12	0C	FF	28	1C	FS	44	2C	,	60	3C	<	76	4C	L	92	5C	\	108	6C	l	124	7C	|
13	0D	CR	29	1D	GS	45	2D	-	61	3D	=	77	4D	M	93	5D	]	109	6D	m	125	7D	}
14	0E	SO	30	1E	RS	46	2E	.	62	3E	>	78	4E	N	94	5E	^	110	6E	n	126	7E	~
15	0F	SI	31	1F	US	47	2F	/	63	3F	?	79	4F	O	95	5F	_	111	6F	o	127	7F	DEL

Each cell is decimal, hex, then the character. 'a' - 'A' = 32 — one bit apart.

And Above 127?

128 to 255: not ASCII, and not one answer

These are the ISO 8859-1 (Latin-1) meanings. Windows-1252 disagrees about 128–159.
In UTF-8 — what your files almost certainly are — none of these bytes is a character on its own.

128	80	ctl	144	90	ctl	160	A0		176	B0	°	192	C0	À	208	D0	Ð	224	E0	à	240	F0	ð
129	81	ctl	145	91	ctl	161	A1	¡	177	B1	±	193	C1	Á	209	D1	Ñ	225	E1	á	241	F1	ñ
130	82	ctl	146	92	ctl	162	A2	¢	178	B2	²	194	C2	Â	210	D2	Ò	226	E2	â	242	F2	ò
131	83	ctl	147	93	ctl	163	A3	£	179	B3	³	195	C3	Ã	211	D3	Ó	227	E3	ã	243	F3	ó
132	84	ctl	148	94	ctl	164	A4	¤	180	B4	´	196	C4	Ä	212	D4	Ô	228	E4	ä	244	F4	ô
133	85	ctl	149	95	ctl	165	A5	¥	181	B5	µ	197	C5	Å	213	D5	Õ	229	E5	å	245	F5	õ
134	86	ctl	150	96	ctl	166	A6	¦	182	B6	¶	198	C6	Æ	214	D6	Ö	230	E6	æ	246	F6	ö
135	87	ctl	151	97	ctl	167	A7	§	183	B7	·	199	C7	Ç	215	D7	×	231	E7	ç	247	F7	÷
136	88	ctl	152	98	ctl	168	A8	¨	184	B8	¸	200	C8	È	216	D8	Ø	232	E8	è	248	F8	ø
137	89	ctl	153	99	ctl	169	A9	©	185	B9	¹	201	C9	É	217	D9	Ù	233	E9	é	249	F9	ù
138	8A	ctl	154	9A	ctl	170	AA	ª	186	BA	º	202	CA	Ê	218	DA	Ú	234	EA	ê	250	FA	ú
139	8B	ctl	155	9B	ctl	171	AB	«	187	BB	»	203	CB	Ë	219	DB	Û	235	EB	ë	251	FB	û
140	8C	ctl	156	9C	ctl	172	AC	¬	188	BC	¼	204	CC	Ì	220	DC	Ü	236	EC	ì	252	FC	ü
141	8D	ctl	157	9D	ctl	173	AD		189	BD	½	205	CD	Í	221	DD	Ý	237	ED	í	253	FD	ý
142	8E	ctl	158	9E	ctl	174	AE	®	190	BE	¾	206	CE	Î	222	DE	Þ	238	EE	î	254	FE	þ
143	8F	ctl	159	9F	ctl	175	AF	¯	191	BF	¿	207	CF	Ï	223	DF	ß	239	EF	ï	255	FF	ÿ

A byte above 127 means nothing until you know the encoding it was written in.
Ñ is 209 here and ñ is 241 — but in UTF-8 each one takes **two** bytes, so "Peña" is 5 bytes, not 4.

ASCII stops at 127 — it only ever needed 7 bits. Everything above that is a *different standard's* idea of what the byte means.

So – Can We Type Spanish?

Put the two tables together and you have 256 slots. Is that enough for every character Spanish actually needs?

Every character Spanish needs has a slot

Blue came from the ASCII table. Orange came from the half above 127.

¿	C	u	á	n	t	o	s	␣	a	ñ	o	s	?
191	67	117	225	110	116	111	115	32	97	241	111	115	63

14 characters, 14 bytes — one byte each, and nothing is missing.

ñ Ñ á é í ó ú Á É Í Ó Ú ü Ü ¿ ¡ — all sixteen live in the upper half.

> **Important**
>
> Yes — but only because someone chose Latin-1. That same byte 241 is ń in Latin-2, Я in KOI8-R and ρ in Greek, and nothing in the file says which. Spanish and Polish cannot share one 8-bit document at all. That is what UTF-8 fixed, at the price that ñ takes two bytes — so `strlen("Peña")` is 5.

How Does C Know What to Print?

The bits never say what they are. `printf` prints whatever the **format code** asks for — so the same `x` comes out as a number or as a character:

```
int x = 65;
printf("%d\n", x);       // 65
printf("%c\n", x);       // A

unsigned a = 12, b = 10;
printf("%u\n", a & b);  // 8
printf("%u\n", a << 2); // 48
```

Nothing about `x` changed between those first two lines. The only thing that changed is what you told `printf` to do with it.

Telling `printf` What You Handed It

`printf` cannot see the type of what you pass — the format string is how it finds out. Get that wrong and the output is garbage, or worse:

Code	Prints
`%d`	a signed `int`
`%u`	an `unsigned`
`%c`	one character
`%s`	a string (`char *`)
`%zu`	a `size_t`, as from `sizeof`

Code	Prints
`%x` `%X`	hex, lower or upper case
`%f`	a `double`
`%p`	a pointer — cast it to `void *`
`%%`	a literal `%` sign
`%02X`	hex, padded to 2 with zeros

⚠ **Warning**

Build with `-Wall` and the compiler catches most mismatches — `%s` with an `int`, `%d` with a `double`. It stays quiet about `%d` with an `unsigned`, though, because they are the same width. That one is still wrong, and it is the one you will write.

Bit-Level Operations

For these bit patterns, C provides six bitwise operators:

Op	Name	Example
`&`	AND	`0b1100 & 0b1010 = 0b1000`
`|`	OR	`0b1100 | 0b1010 = 0b1110`
`^`	XOR(Exclusive OR)	`0b1100 ^ 0b1010 = 0b0110`
`~`	NOT	`~0b1100 = 0b1111...11110011` (every bit flips, including the leading zeros)
`<<`	Left shift	`0b0011 << 2 = 0b1100`
`>>`	Right shift	`0b1100 >> 2 = 0b0011`

Warning

Right shift of signed integers is **implementation-defined** in C. GCC uses **arithmetic** right shift (preserves sign bit). Don't rely on this unless you're sure of your compiler.

Bitwise Operations, One Column at a Time

The same two inputs, three separate rules

Every output bit is decided on its own, from just the two bits directly above it.

INPUTS

a	1	1	0	0	= 1100
b	1	0	1	0	= 1010

a & b	1	0	0	0	AND = 1000 1 only where both are 1

a | b	1	1	1	0	OR = 1110 1 where either is 1

a ^ b	0	1	1	0	XOR = 0110 1 where they differ

De Morgan's Laws

Two rules let you rewrite any mix of &, | and ~ — and they are how Lab 1 asks you to build ^ out of nothing but ~ and &.

Flip the whole thing, and AND becomes OR

`~(x & y) == ~x | ~y`

`~(x | y) == ~x & ~y`

`x`	`y`	`x & y`	`~(x & y)`	`~x | ~y`
0	0	0	1	1
0	1	0	1	1
1	0	0	1	1
1	1	1	0	0

same in all four rows

Two inputs give four possible rows, so checking all four is a complete proof.

Shifting Bits

Shifting slides every bit sideways

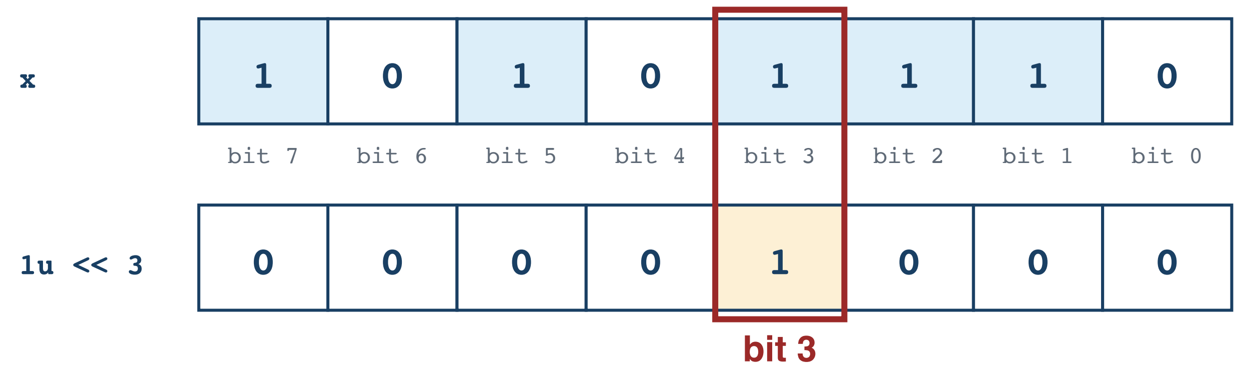
`x`	0	0	0	0	1	1	0	0	= 12
`x << 2`	0	0	1	1	0	0	0	0	= 48 (× 4)
`x >> 2`	0	0	0	0	0	0	1	1	= 3 (÷ 4)

Bits pushed off the end are gone for good; zeros come in behind them.

Shifting left by n multiplies by 2^n. Shifting an unsigned value right by n divides by 2^n.

Test, Set, Clear, Toggle

1u << 3 builds a mask with exactly one bit set



Bit 0 is the rightmost — the ones place. The mask is 0 everywhere except the bit you want.

```c
unsigned x = 0xAE;               // 10101110
x |=  (1u << 3);                 // SET    – bit 3 becomes 1
x &= ~(1u << 3);                 // CLEAR  – bit 3 becomes 0
x ^=  (1u << 3);                 // TOGGLE – bit 3 flips
if (x & (1u << 3)) { }           // TEST   – nonzero when bit 3 is 1
```

Those four are most of Lab 1. Keep the value **unsigned** — shifting into a sign bit is undefined.

What Two's Complement Means

Complement means the part that completes a whole. Flip every bit of x to get its *one's complement*; add 1 to that and you have its **two's complement**, which is how C writes –x:

To write –5 in 8 bits

1. Start with +5: `00000101`

   ↓ flip every 0 ↔ 1

2. One's complement: `11111010`

   ↓ + 1

3. Two's complement: `11111011` = –5

More 8-bit examples

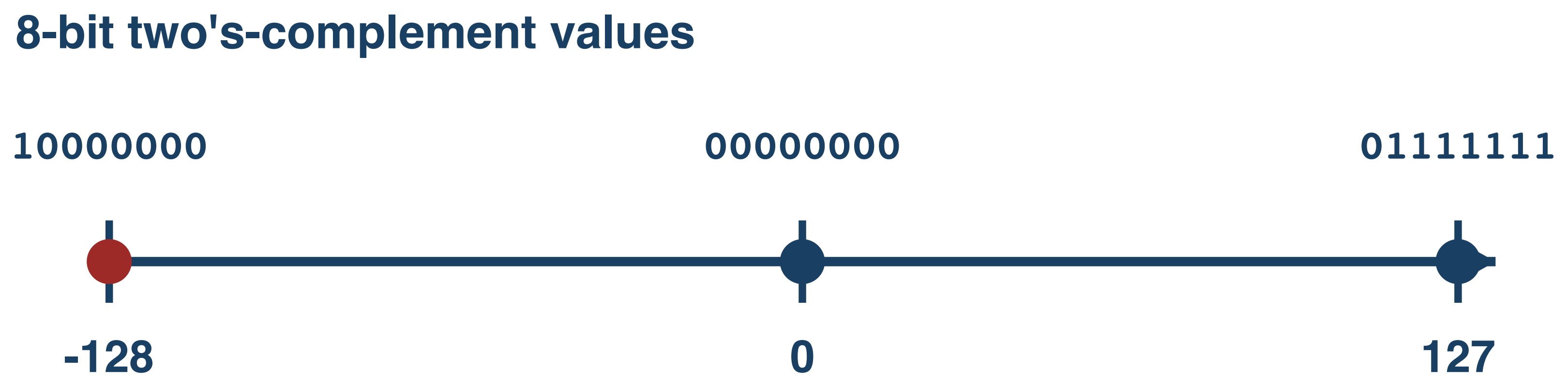
```
+1:  00000001
-1:  11111111

0:  00000000
two's:  00000000
```

Zero stays zero.

A ninth bit is simply discarded — 8 bits keep only their rightmost 8.

The Range It Buys You

An N-bit two's complement covers `−2^(N−1)` to `2^(N−1) − 1`:



There is one more negative value than positive value.

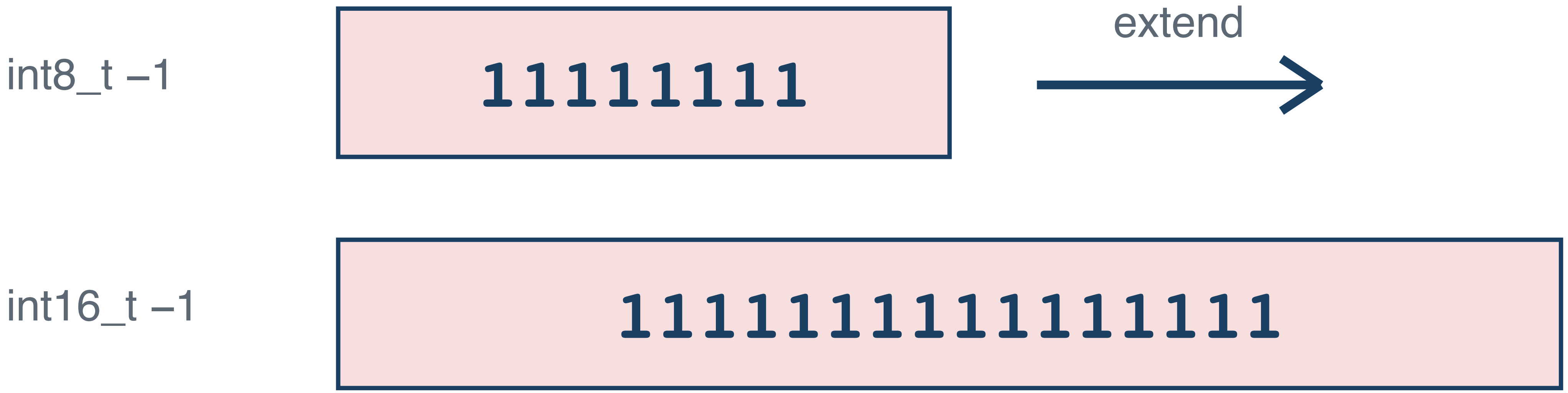
`TMin = −TMax − 1` — there is no positive 128 in 8 bits, so negating `TMin` overflows and is undefined.

Sign Extension

When converting a smaller signed type to a larger one, copy the sign bit:

```
int8_t  x = -1;    // 0b11111111
int16_t y = x;     // 0b1111111111111111 = -1 ✓
```

Copy the sign bit into the new high bits



Truncation

When converting larger to smaller, drop the high bits:

```
int16_t  x = 257;
uint8_t  y = x;    // 1: conversion to unsigned is modulo 256
```

Keep the low bits; discard the high bits

int16_t 257	00000001	00000001

keep these low 8 bits

int8_t result = 00000001 = 1

Size of Integer Types

```c
printf("char:      %zu\n", sizeof(char));      // 1 C byte (not always 8 bits)
printf("short:     %zu\n", sizeof(short));     // 2
printf("int:       %zu\n", sizeof(int));       // 4
printf("long:      %zu\n", sizeof(long));      // 8 (LP64)
printf("long long: %zu\n", sizeof(long long)); // 8
printf("pointer:   %zu\n", sizeof(void*));     // 8
```

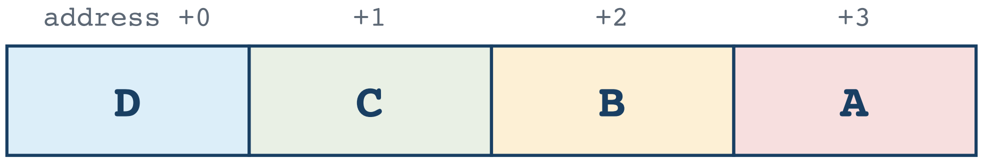
Tip

Use `sizeof` and fixed-width types (`int32_t`, `uint64_t`) when exact sizes matter. Never assume `int` is 4 bytes; use `CHAR_BIT` when bit width matters. Plain `char` is its own trap: whether it is signed is implementation-defined, so say `signed char` or `unsigned char` when it matters.

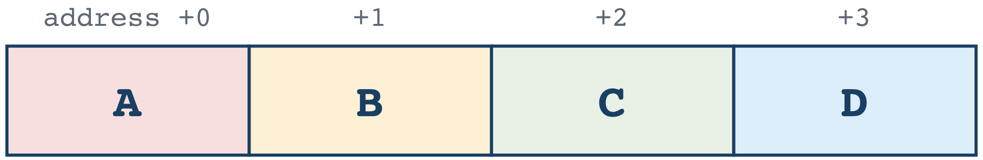
Byte Ordering (Endianness)

An `int` occupies 4 consecutive bytes, so those 4 bytes have to be laid out in some order. **Endianness** is that order. One question matters: **which byte goes at the lowest address?**

Little-endian: D is first in memory



Big-endian: A is first in memory




```c
int x = 1;   // ask the machine: which byte of x sits at the lowest address?
if (*(unsigned char *)&x == 1) printf("Little-endian\n");
else                           printf("Big-endian\n");
```

Bit Tricks in the Wild

LeetCode 136 — Single Number. Every value appears twice except one. Find it in one pass, using no extra memory.

```c
int singleNumber(int *nums, int n) {
    int x = 0;
    for (int i = 0; i < n; i++)
        x ^= nums[i];
    return x;
}
```

```
nums:   5    5    1    2    2

x = 0
  ^ 5  ->   5
  ^ 5  ->   0     the pair cancels
  ^ 1  ->   1
  ^ 2  ->   3
  ^ 2  ->   1     answer
```

`a ^ a` is 0, `a ^ 0` is `a`, and XOR does not care about order — so every pair annihilates wherever the two copies sit, and the unpaired value is what survives. Shuffle the array to `2 5 1 5 2` and the answer is still 1.

AI Systems Connection

Quantization is one important reason large models can fit on more hardware. A common symmetric int8 scheme represents a block of weights with 8-bit signed values and a scale factor:

```
real_value  ≈  scale × int8_value          scale = max_abs / 127
```

Everything today shows up here. Two int8 values need 16 bits for their product, so a long dot product accumulates into something wider — often `int32_t`. And if a signed weight is loaded as *unsigned* into a wider register, sign extension does not happen, negatives become huge positives, and the answer is quietly wrong.

Try it: quantize a float array with `scale = max_abs / 127`, dequantize, and measure the error.

Practice

1. Write `0xD6` in binary, and `10110010` in hex, without using a computer.
2. What does `x & 1` tell you about `x`? And what does `x & (x − 1)` do? Try both on `0b0110` by hand.
3. Write one expression that clears bit 5 of `unsigned x`, and one that tests whether bit 0 and bit 7 are both set.
4. What is `TMax` for a 32-bit `int`? What is `TMin`?
5. Write a C expression that checks if your machine is little-endian.
6. Is left-shifting a 32-bit `int` by 32 positions defined in C? Why not?
7. Why is `−TMin` not representable in the same signed type?

Reading

- CS:APP §2.1 (Information Storage)
- CS:APP §2.2 (Integer Representations)